#define in C

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In C programming, **#define** is a preprocessor directive that is used to define macros. The macros are the identifiers defined by #define which are replaced by their value before compilation. We can define constants and functions like macros using #define. The generics in C are also implemented using the #define preprocessor directive along with _Generic.

Syntax of C #define

The syntax of #define preprocessor directive in C is:

For Defining Constants

```
#define MACRO_NAME value
```

For Defining Expressions

```
#define MACRO_NAME (expression within brackets)
```

For Defining Expression with Parameters

Arguments passed in the macros can be used in the expression.

```
#define MACRO_NAME(ARG1, ARG2,..) (expression within brackets)
```

There are a few more ways using which we can define macros. To know more, refer to this article – Macros and its types in C

Examples of C #define

Example 1:

In the below example, we have defined a macro **'PI'** and assigned it a constant value which we can use later in the program to calculate the area of a circle.

C

```c
// C Program to illustrate how to use #define to declare
// constants
#include <stdio.h>

// Defining macros with constant value
#define PI 3.14159265359

int main()
{

    int radius = 21;
    int area;

    // Using macros to calculate area of circle
    area = PI * radius * radius;

    printf("Area of Circle of radius %d: %d", radius, area);

    return 0;
}
```

Output

```
Area of Circle of radius 21: 1385
```

Example 2:

In the below example, we have defined a macro **'PI'** and assigned it an expression, and that value of the expression is used in the program using **'PI'**.

C

```c
// C Program to illustrate the defining of expression using
// #define
#include <stdio.h>
```

```c
// Defining macros with expression
#define PI (22 / 7)

int main()
{

    int radius = 7;
    int area;

    // Using macros to calculate area of circle
    area = PI * radius * radius;

    printf("Area of Circle of radius %d: %d", radius, area);

    return 0;
}
```

Output

```
Area of Circle of radius 7: 147
```

Example 3:

In the below example, we have defined two macros **CIRCLE_AREA** and **SQUARE_AREA** with a parameter and that parameter is used in the expression to calculate the area of circle and square respectively.

C

```c
// C Program to define the function like macros using
// #define
#include <stdio.h>

// Defining parameterized macros with expression
#define CIRCLE_AREA(r) (3.14 * r * r)
#define SQUARE_AREA(s) (s * s)

int main()
{

    int radius = 21;
    int side = 5;
```

```
    int area;

    // Using macros to calculate areas by
    // passing argument
    area = CIRCLE_AREA(radius);
    printf("Area of Circle of radius %d: %d \n", radius,
           area);

    area = SQUARE_AREA(side);
    printf("Area of square of side %d: %d", side, area);

    return 0;
}
```

Output

```
Area of Circle of radius 21: 1384
Area of square of side 5: 25
```

Important Points

- Macros declared using #define are used to store constants and cannot be changed. we cannot assign variables to the macros.
- We cannot use the **'='** operator to assign value to the macros (eg. **#define PI 3.14**).
- We do not use the semicolon **';'** at the end of the statement in #define.